

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 10

Name:	<u> Muhammad Usman </u>		Roll number:	<u> 23BSCYS006 </u>	
Score:	<u> </u>	Date:	<u> </u>	Instructor's Signature:	<u> </u>

Lab # 10: To configure Load Balancing Among Servers on Different Network Segments

Objective: The objective of this lab is to configure load balancing among servers located on different network segments by utilizing a Virtual Server IP (VSIP) on the firewall, ensuring efficient traffic distribution, high availability, and optimal resource utilization. **Duration: 3 Hours**

Expected Outcomes: By the end of this lab session, students will be able to:

1. Efficient Traffic Distribution
2. High Availability and Reliability:

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in [Figure 1](#), an enterprise has three FTP servers: Server1, Server2, and Server3. The hardware performance of the servers decreases in sequence. Server1 performance is twice of Server2 performance. Server2 performance is twice of Server3 performance. SLB is configured to allow the three servers to provide FTP services as a whole. The carried services match the server hardware performance. In addition, the health check function is configured to monitor the work of the servers.

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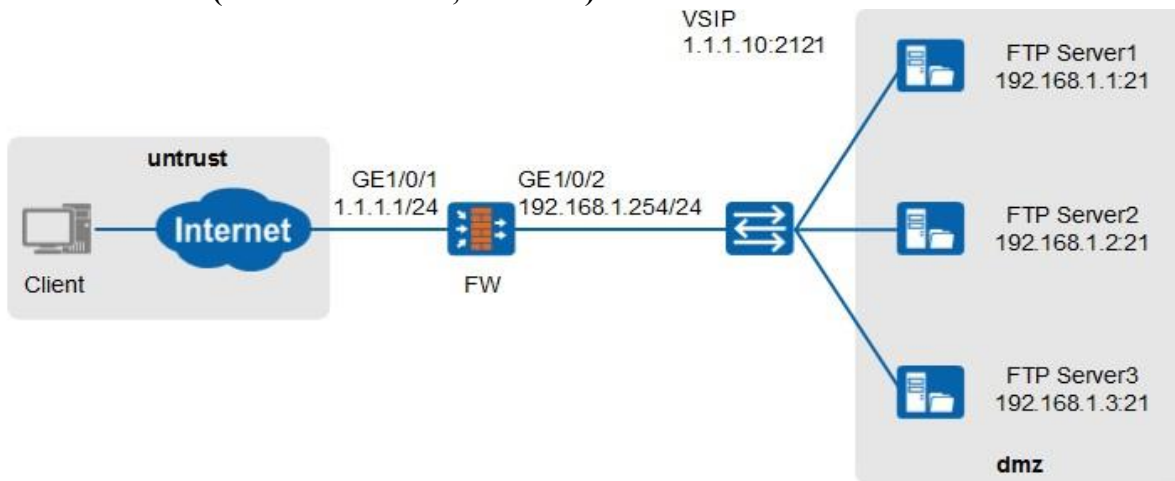


Figure 1 Networking diagram for configuring FTP server load balancing

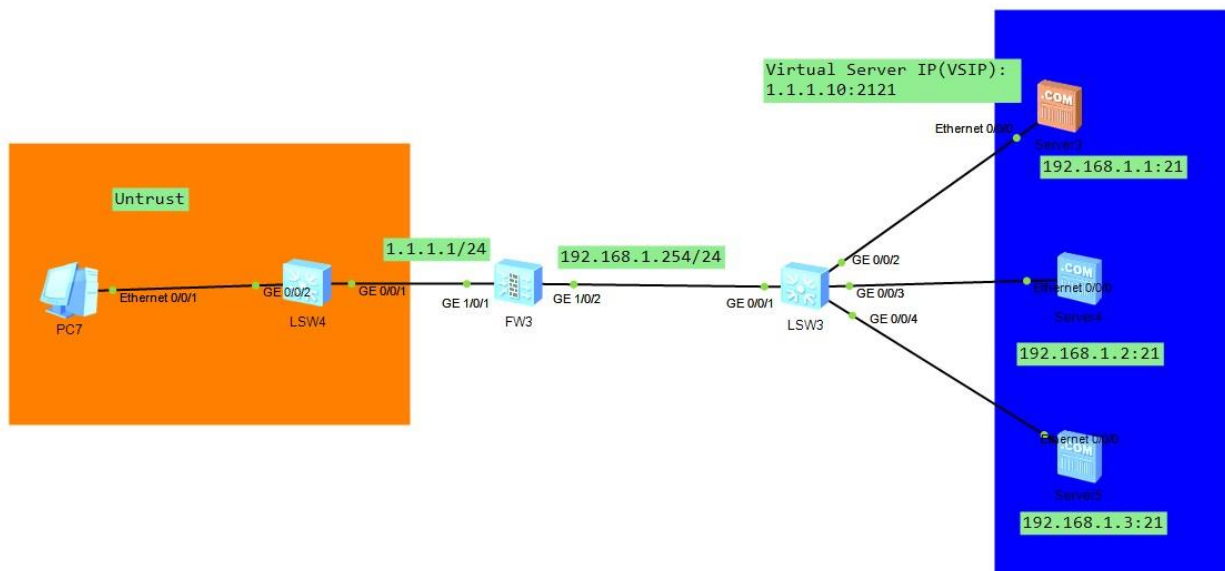


Figure 2 Networking Topology on eNSP

Configuration Roadmap

1. Set the load balancing algorithm to **weight-least-connection** for performancebased SLB. As Server1 performance is twice of Server2 performance and Server2 performance is twice of Server3 performance, the weight ratio of the servers is 4:2:1.
2. Configure the health check function on the FW to check server connectivity. In this example, the protocol type is set to ICMP for health check. You can also use TCP.

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To allow the FW to send health check packets, configure a Local -> DMZ security policy (the servers reside in the DMZ).

3. Configure the sticky session function on the FW because a client may need to establish multiple connections to a server for one task. In this manner, the connection requests from the client can be allocated to the same server. As the servers provide FTP services, the sticky session mode can only be **source-ip**.
4. Enable ASPF for FTP. ASPF is required only for multi-channel protocols.

Procedure

1. Set the IP addresses of the interfaces and add the interfaces to security zones.

```
<FW> system-view
[FW] interface GigabitEthernet 1/0/1
[FW-GigabitEthernet1/0/1] ip address 1.1.1.1 24
[FW-GigabitEthernet1/0/1] quit
[FW] interface GigabitEthernet 1/0/2
[FW-GigabitEthernet1/0/2] ip address 192.168.1.254 24
[FW-GigabitEthernet1/0/2] quit
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/1
[FW-zone-untrust] quit
[FW] firewall zone dmz
[FW-zone-dmz] add interface GigabitEthernet 1/0/2
[FW-zone-dmz] quit
```

2. Configure security policies.

Configure an Untrust -> DMZ security policy to allow Internet users to access FTP servers on the enterprise network. The destination address in the policy should be the IP addresses of real servers, not the IP address of the virtual server.

```
[FW] security-policy
[FW-policy-security] rule name policy1
[FW-policy-security-rule-policy1] source-zone untrust
[FW-policy-security-rule-policy1] destination-zone dmz
[FW-policy-security-rule-policy1] destination-address 1.1.1.10 24
[FW-policy-security-rule-policy1] action permit
[FW-policy-security-rule-policy1] quit
[FW-policy-security] quit
```

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Configure a Local -> DMZ security policy to allow the FW to send health probe packets to the real servers.

```
[FW] security-policy
[FW-policy-security] rule name policy2
[FW-policy-security-rule-policy2] source-zone local
[FW-policy-security-rule-policy2] destination-zone dmz
[FW-policy-security-rule-policy2] destination-address range 192.168.1.1 192.168.1.3
[FW-policy-security-rule-policy2] action permit
[FW-policy-security-rule-policy2] quit
[FW-policy-security] quit
```

3. Configure SLB and health check.

Enable SLB.

```
[FW] slb enable
```

Configure the sticky session function.

```
[FW] slb
[FW-slb] persistence 0 Session
[FW-slb-persistence-1] type cookie-insert aging-time 600
[FW-slb-persistence-1] quit
```

Configure a load balancing algorithm.

```
[FW-slb] group 0 vs-ftp-1.1.1.10
[FW-slb-group-0] metric weight-least-connection
```

Set the weights of Server1, Server2, and Server3 to 4, 2, and 1 separately.

```
[FW-slb-group-0] rserver 0 rip 192.168.1.1 port 21 weight 4 description server1
[FW-slb-group-0] rserver 1 rip 192.168.1.2 port 21 weight 2 description server2
[FW-slb-group-0] rserver 2 rip 192.168.1.3 port 21 weight 1 description server3
```

Configure the health check function.

```
[FW-slb-group-0] health-check type icmp tx-interval 5 times 3
[FW-slb-group-0] quit
```

Configure a protocol type for the virtual server.

```
[FW-slb] vserver 0 vs-ftp-1.1.1.10
[FW-slb-vserver-0] protocol tcp
```

Assign an IP address and a port number to the virtual server.

```
[FW-slb-vserver-0] vip 0 1.1.1.10
[FW-slb-vserver-0] vport 2121
```

Associate the virtual server with the real server group.

```
[FW-slb-vserver-0] group vs-ftp-1.1.1.10
```

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```
[FW-slb-vserver-0] quit
```

4. Enable ASPF for FTP.

```
[FW] firewall detect ftp
```

Verification

1. Check static server map entries on the FW. If corresponding server map entries exist, the server load balancing is successfully configured.

```
[FW] display firewall server-map static
Current Total Server-map : 1
Type: SLB, ANY -> 1.1.1.10:2121[vs-ftp-1.1.1.10/0], Zone:--, protocol:tcp
Vpn: public -> public
```

Task:

Attach Screenshots of all the topology and configuration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-23BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. Why is weight-least-connection used and what is the weight ratio?

The **weight-least-connection** algorithm is used to distribute traffic according to server performance.

More powerful servers handle more connections.

The weight ratio is:

- **Server1 = 4**
- **Server2 = 2**
- **Server3 = 1**

So the ratio is **4 : 2 : 1**.

2. What is the purpose of health check and which protocols are used?

The health check is used to make sure the servers are **alive and working**.

If a server goes down, the firewall stops sending traffic to it.

Protocols used for health check in this lab:

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- **ICMP (Ping)**
- **TCP**

3. Why is Local → DMZ security policy required?

The firewall sends health check packets from itself (Local zone) to the servers (DMZ zone). Without a **Local → DMZ policy**, the firewall would block its own health check traffic. So this policy allows the firewall to check server status.

4. What is the role of sticky session and why source-ip mode?

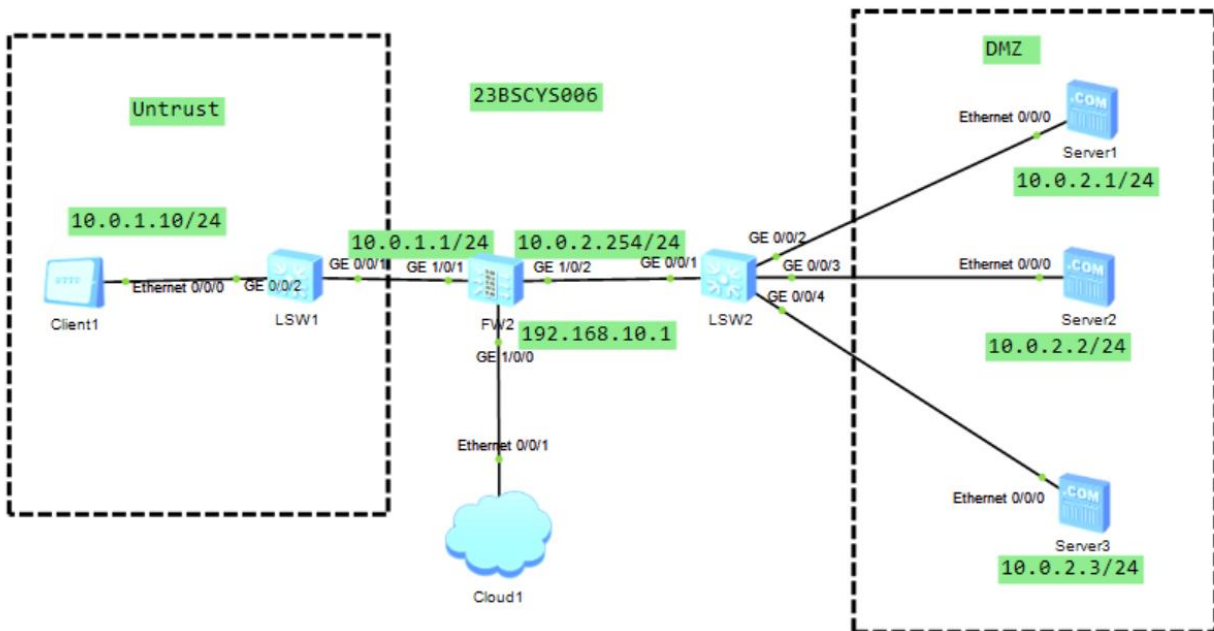
Sticky session ensures that a client always connects to the **same server** during a session. FTP opens multiple connections, so they must go to the same server. The mode is set to **source-ip** because FTP servers do not support cookie-based persistence.

5. Why is ASPF required and which protocols need it?

ASPF is required because FTP is a **multi-channel protocol** (control and data connections). ASPF helps the firewall understand and allow these extra connections. ASPF is needed for:

- **Multi-channel protocols** like FTP.

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Security Policy List

[Add Security Policy](#)
[Add Security Policy Group](#)
[Delete](#)
[Copy](#)
[Move](#)
[Insert](#)
[Export](#)
[Reset All Statistics](#)
[Enable](#)
[Disable](#)
[Customize](#)
[Expand](#)
[Collapse](#)

[Refresh](#)
[Match Query](#)
[Clear Match](#)

Enter a keyword. [Add Filter](#)

Index	Name	Descri...	Tag	VLAN ID	Sourc...	Destin...	Sourc...	Destin...	User	Service	Applic...	Sched...	Action	Conte...	Hits	Enable	Edit
1	Local-To-DMZ			any	local	dmz	any	Ser...	any	any	any	any	Permit		0	Reset	☒
2	Untrust-To-D...			any	untrust	dmz	Unt...	any	any	any	any	any	Permit		11	Re...	☒
3	Trust-To-DMZ			any	trust	dmz	My...	any	any	any	any	any	Permit		0	Reset	☒
4	default	This is...		any	any	any	any	any	any	any	any	any	Deny		6	Reset	☒

[Add](#)
[Delete](#)

[Refresh](#)

[Search](#)
[Clear Search Condition](#)

Name	Type	Timeout	Hold Mode	Cookie Name	Edit
☒ persistence0Session	Cookie	600	Insert		Edit

Real Server Group List

[Add](#)
[Delete](#)

[Refresh](#)

[Search](#)
[Clear Search Condition](#)

Name	Load Balan...	Health Chec...	Real Server ...	Weight	Health Status	Health Status ...	5-Minute Statistics			Concurrent ...	Edit
							Traffic	Traffic Percentage	Sessions		
☐ group0	Weighted Le...	ICMP	10.0.2.1:21	4(57.14%)		1 Reset	5.18945KB		79.55%	4	4
			10.0.2.2:21	2(28.57%)		1 Reset	1.33398KB		20.45%	1	2
			10.0.2.3:21	1(14.29%)		1 Reset	0B		0.00%	---	---

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Virtual Service List

+ Add - Delete

Name
vserver0vs-ftp-1...

Modify Virtual Service

Name:

Protocol:

Virtual Server IP Address:

Port: <1-65535>

Sticky Session: [Configure]

Real Server Group: [Configure]

Max. Concurrent Connections: <1-4294967295>

Real Server G... Max...

group0

ARP Table

IP-MAC Binding Refresh All Search

IP Address	MAC Address	Aging Time	Type	Interface
☐ 192.168.10.10	0200-4c4f-4f50	20	Dynamic	GE1/0/0
☐ 10.0.1.10	5489-9824-61e6	1	Dynamic	GE1/0/1(WAN)
☐ 10.0.2.3	5489-9860-1f9b	20	Dynamic	GE1/0/2(DMZ)
☒ 10.0.2.1	5489-988b-4d6b	20	Dynamic	GE1/0/2(DMZ)
☐ 10.0.2.2	5489-98a6-51ff	20	Dynamic	GE1/0/2(DMZ)
☐ 192.168.10.1	00e0-fc3b-0879		Interface	GE1/0/0
☐ 10.0.1.254	00e0-fc3b-087a		Interface	GE1/0/1(WAN)
☐ 10.0.2.254	00e0-fc3b-087b		Interface	GE1/0/2(DMZ)

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ASPF Configuration

ASPF allows the transmission of multi-channel protocol packets. It enables the system to implement packet filtering and NAT on these packets based on the negotiated ports.

- | | | | |
|----------------------------------|---|--------------------------------|----------------------------------|
| ☐ DNS | ☒ FTP | ☐ H.323 | ☐ ICQ |
| ☐ ILS | ☐ MGCP | ☐ MMS | ☐ MSN |
| ☐ NetBIOS | ☐ PPTP | ☐ QQ | ☐ RSH |
| ☐ RTSP | ☐ SCCP | ☐ SIP | ☐ SQL*Net |

Apply

Real Server Group List

+ Add - Delete

Refresh Search

☐ Name	Load Balan...	Health Chec...	Real Server ...	Weight	Health Status	Health Status ...	5-Minute Statistics			Concur...
							Traffic	Traffic Percentage	Sessions	
☐ group0	Weighted Le...	ICMP	10.0.2.1:21	4(57.14%)	🔴	2 Reset	0B	0.00%	---	---
			10.0.2.2:21	2(28.57%)	🟢	1 Reset	220B	13.87%	---	2
			10.0.2.3:21	1(14.29%)	🟢	1 Reset	1.33398KB	86.13%	1	1